

A close-up photograph of two hands, palms facing each other, resting on a wooden surface. The fingers and parts of the palms exhibit a distinct bluish-purple discoloration, a clinical sign known as cyanosis, which is characteristic of hypoxemia or certain connective tissue diseases. The background is a warm-toned wooden table.

My fingers turn white and blue

Other connective tissue disorders and
vasculitic syndromes

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18 November 2023

Lecture outline

- Case illustration – common features of connective tissue disorders
- Focus on Raynaud's phenomenon
- Systemic sclerosis
- Sjögren's syndrome
- Dermato- and polymyositis
- Juvenile arthritides
- Primary vasculitides (optional)

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Mrs. S Ho, F/40, housewife

- Previously well
- Complained of 6-months history of “cold fingers”, particularly during winter months
- On exposure to cold, the fingers turned white then blue then red

Mrs. S Ho, F/40, housewife

- Further enquiry revealed
 - Felt tired easily
 - Weight loss of 2 kg over 3 months
 - ‘Swollen’ sensation of both hands
 - Intermittent arthralgia
 - Progressive shortness of breath on exertion

Mrs. S Ho, F/40, housewife

- Investigations
 - CXR - bilateral lower zone haziness with reticular shadows
 - Hgb - 10.9 g/dl (NcNc), ESR - 56 mm/hr
 - SGOT 340 IU/ml; CPK - 980 IU/ml; LDH 896 IU/ml
 - EMG - inflammatory myopathy
 - ANA 1/1080, anti-RNP +ve

Mrs. S Ho, F/40, housewife

- Investigations
 - Lung function tests
 - $\downarrow$ VC and TLC
 - $\uparrow$ FEV₁/VC and RV/TLC
 - $\downarrow$ DLCO
 - HRCT thorax - early fibrosing alveolitis

Mrs. S Ho, F/40, housewife

- Diagnosis is mixed connective tissue disorder (MCTD) complicated by
 - Raynaud's phenomenon
 - Sclerodermatous skin changes
 - Fibrosing alveolitis
 - Low grade myositis

Mrs. S Ho, F/40, housewife

- Salient features
 - Young / middle aged female
 - Subacute onset
 - Progressive course
 - Cold induced vasospasm
 - Multisystem involvement

Common early features of CTD

- Clinical
 - Unexplained fever
 - Polyarthritits
 - Photosensitive rashes
 - Lower limb petechiae
 - Raynaud's
 - Nail fold changes
 - Sicca symptoms
 - Alopecia
 - Recurrent oral ulcers
 - Serositis
- Laboratory
 - Cytopenia
 - High ESR
 - Reversed A/G ratio
 - Urine sediments and protein
 - Positive ANA
 - Hyperglobulinaemia

The spectrum of connective tissue disorders



Predominant fibrosis

Systemic sclerosis

- Raynaud's phenomenon
- Severe scleroderma
- Internal organ involvement
- Anti-scl 70, anti-centromere

uCTD / MCTD

- Raynaud's
- Scleroderma
- Inflammatory rashes
- Inflammatory myositis
- Oesophageal involvement
- Lung involvement
- Anti-RNP

Predominant inflammation

SLE

- Inflammatory rashes
- Inflammatory arthritis
- Immune complex disease
- Multi-system involvement
- Multi-autoantibodies

The spectrum of connective tissue disorders



Focal inflammatory responses

Systemic inflammatory responses

RA
PBC

Sjögren's
uCTD / MCTD
Pm
Dm

SLE

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RP vs acrocyanosis



Raynaud's phenomenon

- Primary
- Secondary
 - **Connective tissue disorders including vasculitis**
 - Vibration white finger syndrome
 - Atherosclerosis
 - Hyperviscosity - cold haemagglutinins, cryoglobulinaemia
 - Hypothyroidism
 - Anaemia (more often exacerbates rather than precipitates RP)
 - Thoracic outlet syndrome
 - Drug induced - e.g. β -blockers, bleomycin

Connective tissue disease and Raynaud's phenomenon

- Systemic sclerosis
- MCTD
- Undifferentiated CTD
- Sjögren's syndrome
- Poly-/dermato-myositis
- Systemic lupus erythematosus
- Necrotising vasculitis e.g. polyarteritis nodosa
- Rheumatoid arthritis

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Systemic sclerosis - definition

- A generalised disorder of connective tissue affecting skin and internal organs.
- Characterised by fibrotic arteriosclerosis of peripheral and visceral vasculature.
- Variable degrees of extracellular matrix accumulation (mainly collagen) occur in both skin and viscera.
- Associated with specific autoantibodies, most notably anticentromere and anti-Scl-70.

Systemic sclerosis - epidemiology

- Peak age of onset – 30–50 years
- Female : Male = 4 : 1
- Prevalence – 10-20 per 100,000
- Annual incidence – 1-2 per 100,000
- Geography – unrestricted
- Genetic predisposition - ?DR5, DRw52, DR4

Systemic sclerosis – clinical syndromes

- Limited SSc

- Skin involvement confined to areas of the extremities below the elbows and below the knees
- Less extensive systemic disease
- Comparatively better prognosis
- Associated with anti-centromere antibody
- Previously known as CREST syndrome

- Diffuse SSc

- Skin involvement of the proximal extremities (above elbow) or the trunk below the clavicles
- Higher risk for major organ involvement, esp pulmonary fibrosis and renal crisis
- Poor prognosis
- Associated with anti-scl 70 and anti-RNA-polymerase III antibodies

Limited SSc - CREST syndrome

- C = Calcinosis cutis
 - R = Raynaud's phenomenon
 - E = oEsophageal dysmotility
 - S = Sclerodactyly
 - T = Telangiectasia
- Associated with anti-centromere antibodies

Diffuse SSc - clinical features

- Cold induced vaso- (RP) / veno- constriction



Severe acrocyanosis



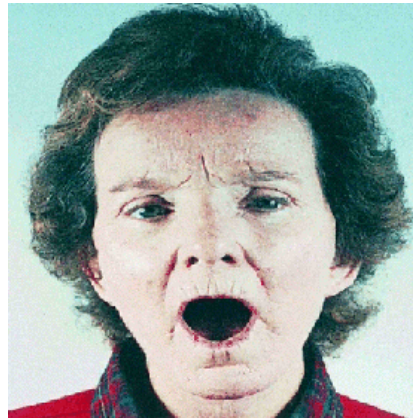
Digital ischaemia



Digital gangrene

Diffuse SSc - clinical features

- Cutaneous involvement
 - Skin tightening and calcinosis cutis



Diffuse SSc - clinical features

- Gastrointestinal involvement
 - Oesophageal reflux → oesophagitis, stricture and bleeding, nocturnal aspiration
 - Small bowel involvement → bloating, abdominal cramps, intermittent or chronic diarrhoea, intestinal obstruction and pseudo-obstruction, malabsorption
 - Large bowel involvement → constipation, diverticular disease



Diffuse SSc - clinical features

- Musculoskeletal involvement
 - Polyarthrititis; 20-30% of patients with erosion
 - Digital tuft resorption
 - Low grade myositis
- Cardio-pulmonary involvement
 - Dysrhythmia and cardiac failure
 - Progressive fibrotic lung disease
 - Pulmonary hypertension
- Renal involvement
 - Accelerated hypertension and renal failure



Systemic sclerosis - Treatment

- **“No drug has been proven to be totally useless until it has been tried in systemic sclerosis”**
- Treatment is largely symptomatic
- Potential disease remitting drugs
 - Endothelial modulators e.g. PGI₂, tPA
 - Immune modulators for inflammatory complications e.g. methotrexate, mycophenolate, cyclosporin A, tacrolimus (watch out for renal toxicity)
 - Anti-fibrotic agents with limited evidence e.g. penicillamine, colchicine, γ -interferon, anti-TGF β
 - Specific inhibitors – endothelin receptor antagonists

Systemic sclerosis - Treatment

- **“Some drugs may not be as useless in systemic sclerosis”**
- Systemic sclerosis associated interstitial lung disease
 - Systemic corticosteroids and cyclophosphamide / mycophenolate mofetil
 - Nintedanib – tyrosine kinase inhibitor
 - Tocilizumab – Anti-IL6 receptor monoclonal Ab

Systemic sclerosis - Treatment

- Raynaud's phenomenon
 - Keep warm; avoid cigarettes and vasoconstrictors; vasodilators (calcium channel blockers, PGI₂ / analogues / phosphodiesterase inhibitors)
- Cutaneous involvement
 - Penicillamine; colchicine; γ -interferon – None has been proven
- Gastrointestinal involvement
 - H₂ blockers; proton pump inhibitor; drugs that improve gut motility
 - Antibiotics for bowel flora overgrowth

Systemic sclerosis - Treatment

- Musculoskeletal involvement
 - Polyarthrititis – NSAID; anti-malarials
 - Low grade myositis - low dose prednisolone
- Cardio-pulmonary involvement
 - Dysrhythmia - anti-arrhythmic drugs
 - Lung fibrosis - steroid and immunosuppressive for early disease
 - Pulmonary hypertension – vasodilators as per for RP; O2, endothelin-1 antagonists, phosphodiesterase 5 inhibitors
- Renal involvement
 - Accelerated hypertension – ACEI

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- Primary vasculitides (optional)

Sjögren's syndrome

- Definition

- A slowly progressive, inflammatory autoimmune disease affecting primarily the exocrine glands
 - Keratoconjunctivitis sicca and xerostomia
- Lymphocytic infiltrates replace functional epithelium leading to decreased exocrine secretions (exocrinopathy)
- Characteristic autoantibodies, Ro(SS-A) and La(SS-B) and rheumatoid factor, are produced

Sjögren's syndrome - epidemiology

- Occurs in patients of all ages
 - 40 - 50 years; Female : male = 9 : 1
- Prevalence is unknown
 - Primary Sjögren's syndrome may be as common as SLE
 - Secondary Sjögren's syndrome = immune mediated keratoconjunctivitis sicca complex + an underlying CTD
 - ~ 30% of RA and SSc have at least histologic evidence of Sjögren's
 - ~2-3% of patients with CTD have focal lymphocytic infiltrates of the labial minor salivary glands

Sjögren's syndrome - clinical features

- Disease of the exocrine glands
 - Mucosal dryness
 - Keratoconjunctivitis sicca
 - Xerostomia
 - Dental caries and gingival diseases
 - Swallowing difficulty
 - Xerotrachea
 - Frequent URTi's
 - Vaginal dryness
 - Dysparenia and vaginal infections
 - Major salivary gland enlargement
 - Atrophic gastritis and pancreatitis



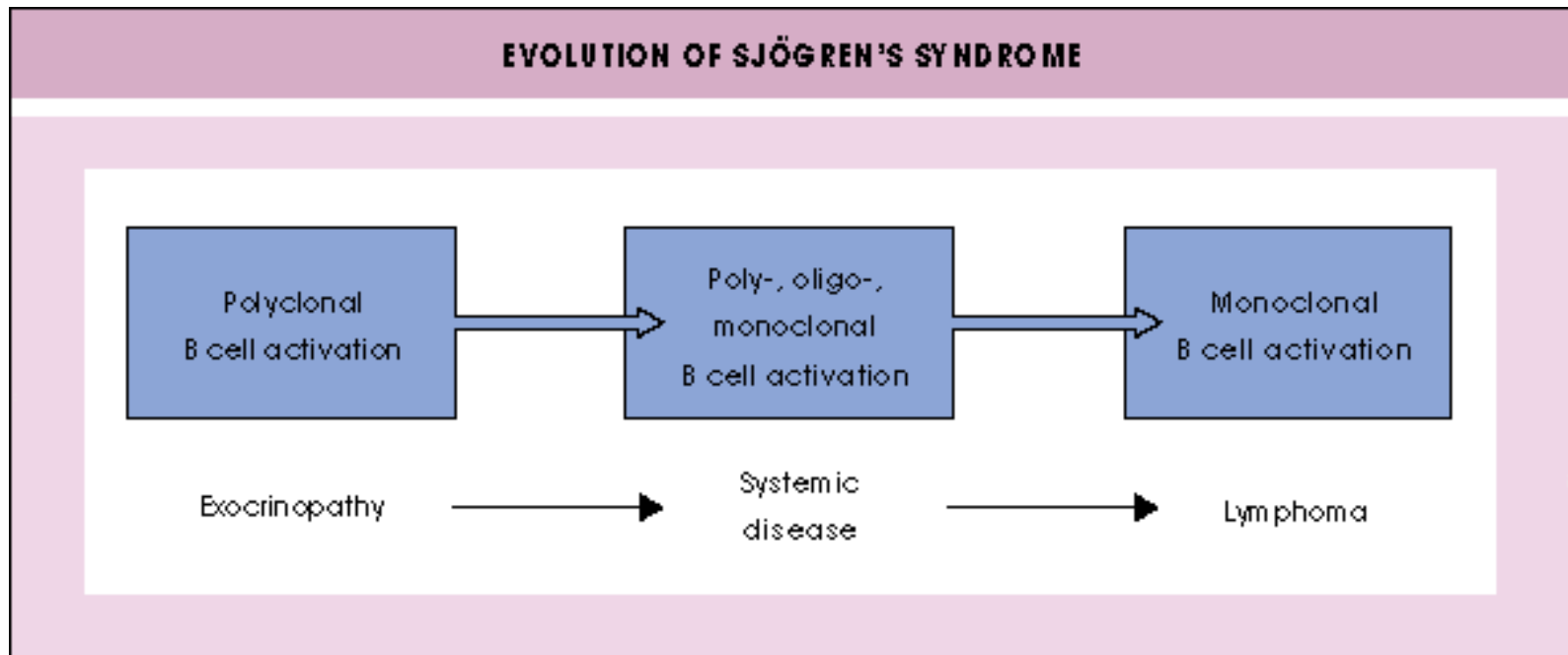
Sjögren's syndrome - clinical features

- Features of autoimmunity
 - Auto-antibodies, cytopenia, glomerulonephritis, etc
- Extra-glandular disease
 - Non-erosive polyarthrititis
 - Raynaud's without telangiectasia / digital ulceration
 - Lung involvement - bronchitis, interstitial lung disease
 - Kidneys - tubular acidosis and glomerulonephritis
 - Blood vessels - vasculitis
 - Muscles - low grade myositis



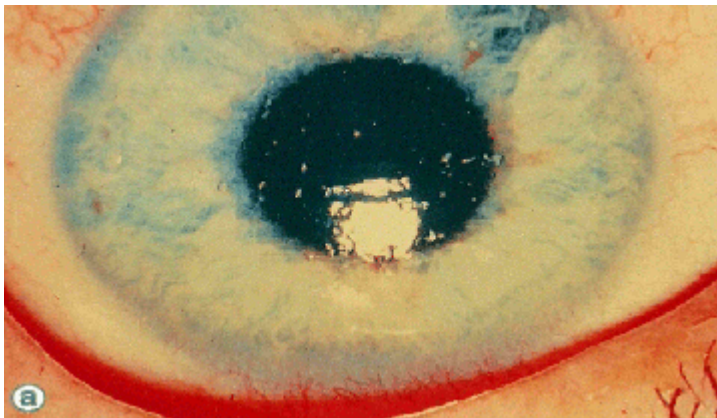
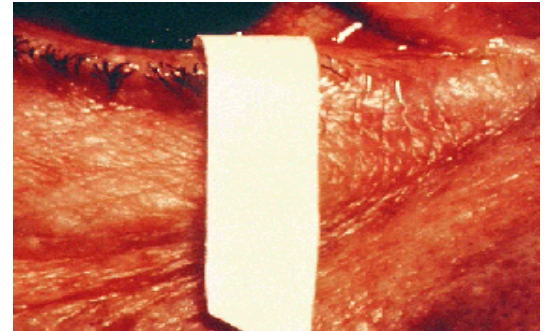
Sjögren's syndrome - clinical features

- Risk of lymphoid malignancy – usually low grade lymphoma

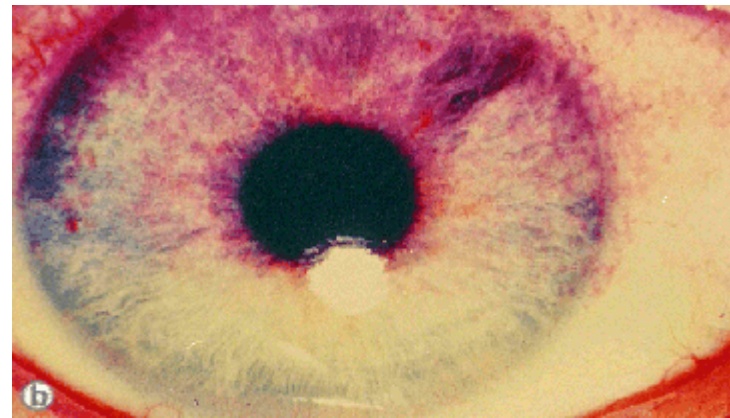


Sjögren's - Investigations

- Demonstration of mucosal dryness
 - Schirmer's test
 - Rose Bengal staining
 - Slit lamp examination
 - Sialography and sialometry



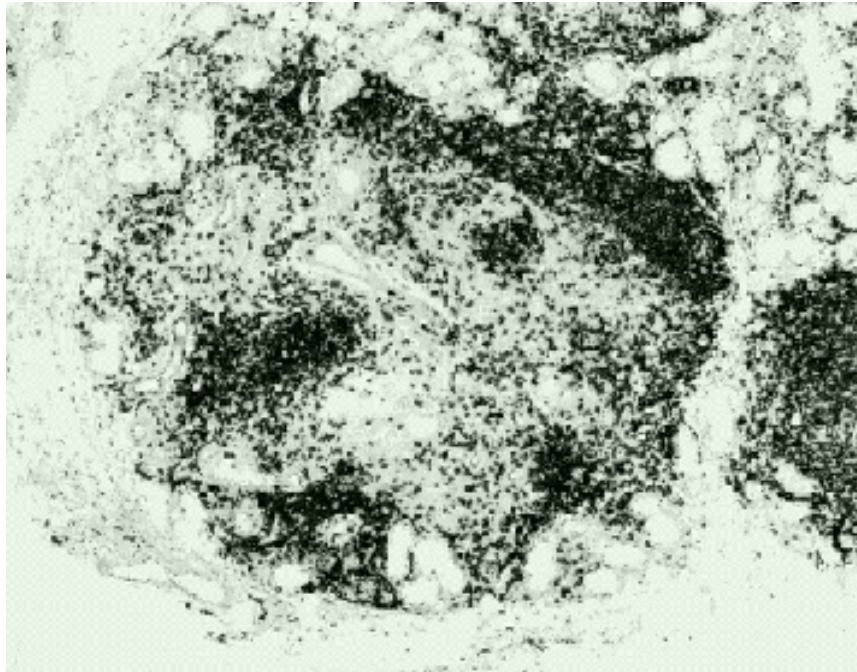
Normal



Sjögren's

Sjögren's - Investigations

- Demonstration of lymphocytic infiltration
 - Major or minor parotid gland swelling



Sjögren's - Investigations

- Demonstration of autoimmunity
 - Hyperglobulinaemia
 - High Ig's
 - Positive rheumatoid factor (>90%)
 - Positive anti-Ro (60%) and La (40%)
 - Leucopenia
 - Thrombocytopenia

Sjögren's - Treatment

- Mucosal dryness
 - Artificial tears and saliva
 - Cholinergic agonists for dry mouth: cevimeline
 - Ensure dental hygiene
 - Prompt treatment of superficial infections
 - Vaginal lubricants
- Systemic illness – as per SLE treatment
 - Steroids (0.5–1mg/kg/day of prednisone) and immunosuppressive drugs such as cyclophosphamide for interstitial pneumonitis, glomerulonephritis, vasculitis and peripheral neuropathy
 - Treatment of lymphoma depends on the histologic type

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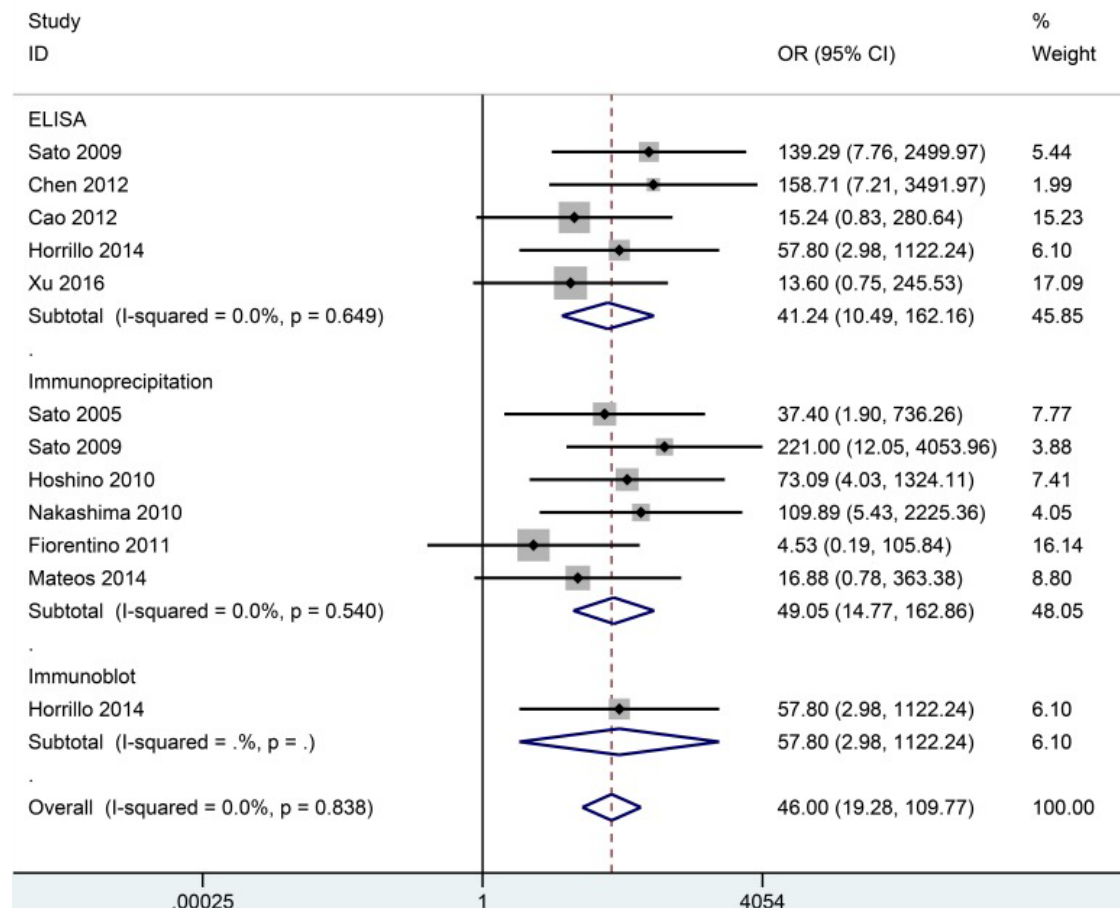
Dermato- and poly-myositis

- Definition

- A member of the connective tissue disease family as evidenced by autoimmune disease associations and other immunologic features
- Characterised by chronic inflammation of striated muscles (polymyositis) and sometimes the skin (dermatomyositis)
- Autoantibody associations (e.g. anti-Jo-1, anti-Mi-2) define homogeneous clinical subsets of disease

Anti-MDA5 antibodies

Anti-MDA5 is also associated with rapid development of interstitial lung disease and high mortality



Forest plot of the association between the presence of anti-MDA5 and clinically amyopathic dermatomyositis

Dermato- and poly-myositis

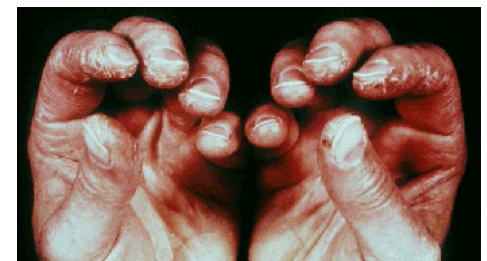
- May present with skin manifestation alone (Dm sine myositis or amyopathic dermatomyositis); myositis alone (Pm); or both (Dm)
 - Childhood idiopathic disease
 - Adult idiopathic disease
 - Associated with other connective tissue disorders
 - Associated with underlying malignancy

Dermatomyositis - clinical features

- Painless proximal muscle weakness with or without rash is the hallmark feature
- Elevation of serum muscle enzymes, most notably creatine kinase
 - Maybe acute or insidious onset
 - Shoulder and pelvic girdle involvement common
 - Neck muscle weakness
 - Bulbar muscle involvement has poor prognosis

Dermatomyositis - clinical features

- Cutaneous involvement
 - Photosensitive rash
 - Heliotrope rash
 - Gottron's patches
 - Calcinosis cutis
 - Digital vasculitis
 - Machinist's hands
 - Raynaud's phenomenon



Dermatomyositis - clinical features

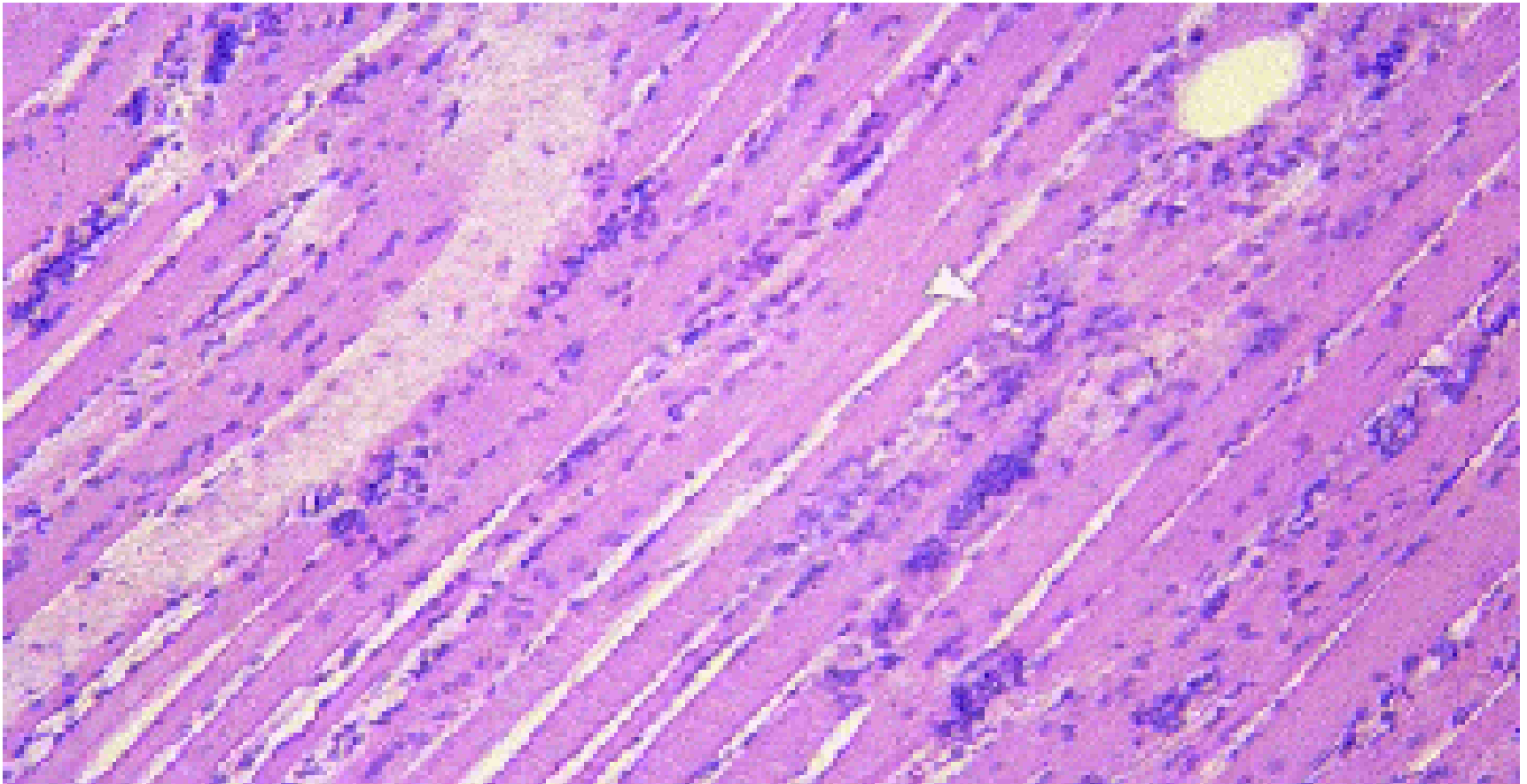
- Musculoskeletal involvement
 - Polyarthrititis
- Pulmonary involvement
 - Lung fibrosis - association with anti-Jo1 antibody
- Probable association with malignancy in elderly population
 - Carcinoma of the nasopharynx (HK), bronchus, breast, stomach, ovary, cervix, prostate



Dm - Investigations

- Typical clinical features provide most clues
- Skin biopsy may sometimes be needed
- Myositis – 2 out of 3 of the following
 - Elevated muscle enzymes - CPK, LDH, AST
 - Electromyography
 - Muscle biopsy
- Auto-Ab – see above
- Thorough search for malignancies is mandatory especially in the elderly

Muscle biopsy



Dermatomyositis - Treatment

- Myositis
 - High dose steroid and other immunosuppressive drugs e.g. azathioprine, methotrexate, cyclophosphamide
 - Intravenous immunoglobulin-G for refractory diseases
 - Anti-CD20
 - Physio- and speech therapy
 - Percutaneous gastrostomy

Dermatomyositis - Treatment

- Cutaneous involvement
 - Moderate to high dose steroid and immunosuppressives
 - Hydroxychloroquine
 - Topical steroids
 - Vasodilators for Raynaud's

Dermatomyositis - Treatment

- Musculoskeletal involvement
 - Polyarthrititis - NSAID
- Pulmonary involvement
 - Lung fibrosis - High dose steroid and immunosuppressives
- Probable association with malignancy in elderly population
 - Thorough search for an underlying malignancy and treat as appropriate

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Juvenile arthritis

Same disease, different names

- Juvenile chronic arthritis - JCA
 - European League of Associations for Rheumatology (EULAR)
- Juvenile rheumatoid arthritis - JRA
 - American College of Rheumatology (ACR)
- Juvenile idiopathic arthritis - JIA
 - International League of Associations for Rheumatology (ILAR)

Types of JIA

- Polyarticular (5 joints or more)
 - Rheumatoid factor +ve
 - RF -ve
- Pauci-articular (4 joints or less)
 - Type 1: Girls > boys
 - Associated with ANA iritis
 - Type 2: Boys > girls
 - Associated with HLA B27
- Systemic onset

Polyarticular onset juvenile idiopathic arthritis

- a subset of juvenile idiopathic arthritis (JIA) that is defined by the presence of more than four affected joints during the first six months of illness
- This subset comprises 30 to 40 percent of patients with JIA
- polyarticular JIA was subclassified by the presence or absence of rheumatoid factor (RF)

Clinical Features

- More frequent in females than males
- Clear bimodal distribution:
 - the first peak in incidence is between the ages of two and five years, and
 - the second peak between ten and fourteen years

Children <10 years of age

- Symmetrical joint involvement is common, with the knees, wrists, and ankles most frequently affected
- Uveitis can occur, although it is much less common than in children with pauciarticular JIA. Those patients with antinuclear antibodies (ANAs) are at greater risk for uveitis than those with a negative ANA

Children >10 years of age

- RF positive children
 - Follow the pattern of gradual disease progression commonly seen in adults with rheumatoid arthritis
- The larger group of children with polyarticular onset disease are RF negative

Pauciarticular onset juvenile idiopathic arthritis (JIA)

- Involvement of fewer than 5 joints
- Females > males
- Peak incidence: second and third years
- Larger joints are affected (knees, ankles, wrists, elbows) but virtually never begins in the hips
- Systemic manifestations (other than uveitis) are characteristically absent. Thus, fever, rash or other constitutional symptoms suggest other diagnosis

Prognosis

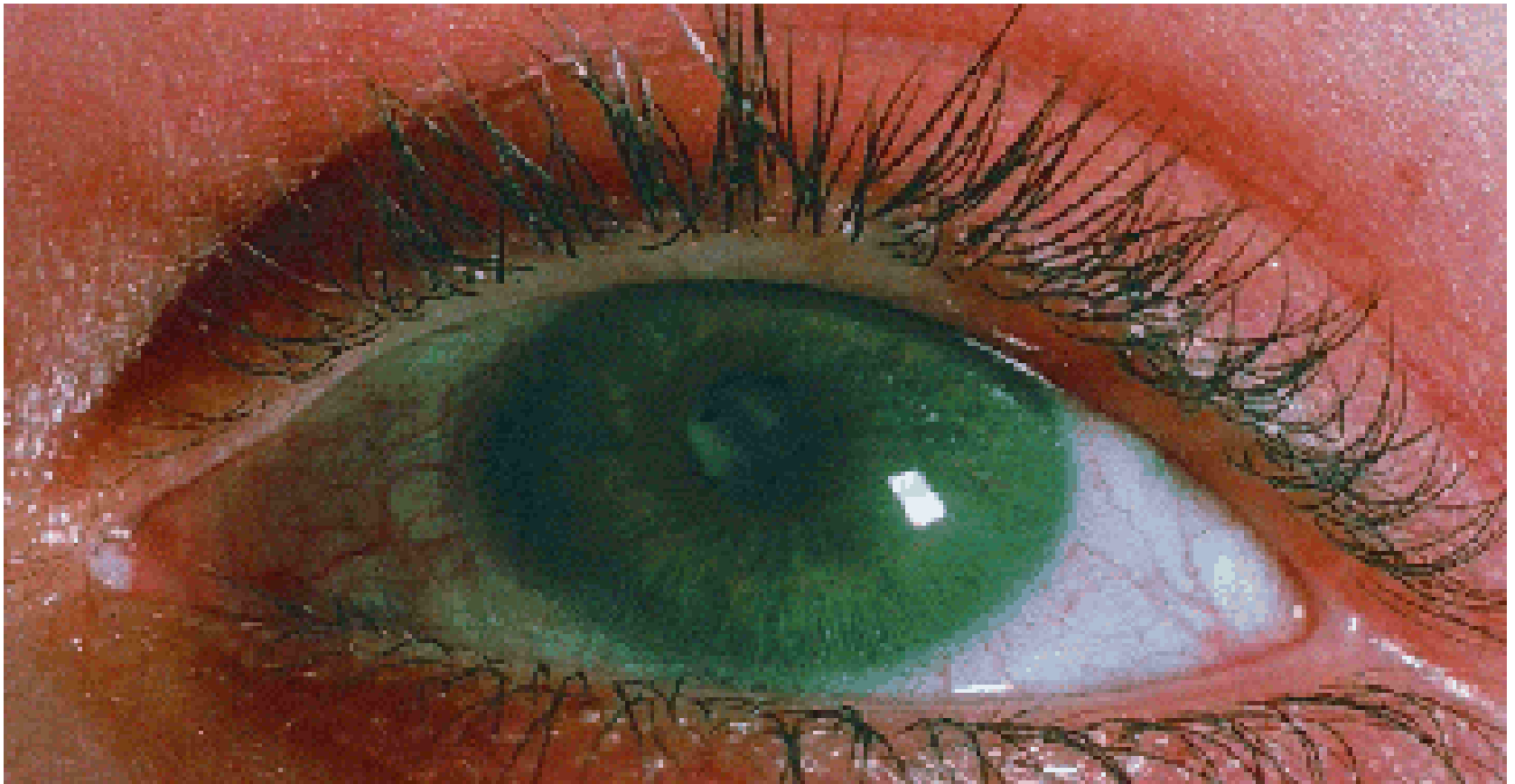
- Several long term complications occur in pauciarticular JIA. The most frequent complications are uveitis and leg length discrepancy

Pauci-articular JIA in the right ankle





Cataract and irregular pupil due to previous iritis



Systemic onset JIA

- Most difficult to diagnose
 - Arthritis, although necessary for a definitive diagnosis, may not be evident early in the course of the disease
 - Constitutional symptoms +++
 - Ill-looking, spiking fever, rashes, high WBC, anaemia
 - Infection is a major differential diagnosis
 - Leukaemia is another possible differential diagnosis

Articular manifestations

- Any number of joints may be involved
 - Wrists, knees and ankles most typical
 - But hands, hips, cervical spine and TM joints may also be affected
- Micrognathia and cervical spine fusion are commonly present in children with chronic disease



Cervical spine fusion in JRA Fusion of the cervical spine may affect up to 50 percent of children with systemic onset JRA in the first decade of life. Apophyseal joint space narrowing and bony ankylosis occur, especially at C2-C3. These changes are not usually present elsewhere in the spine.

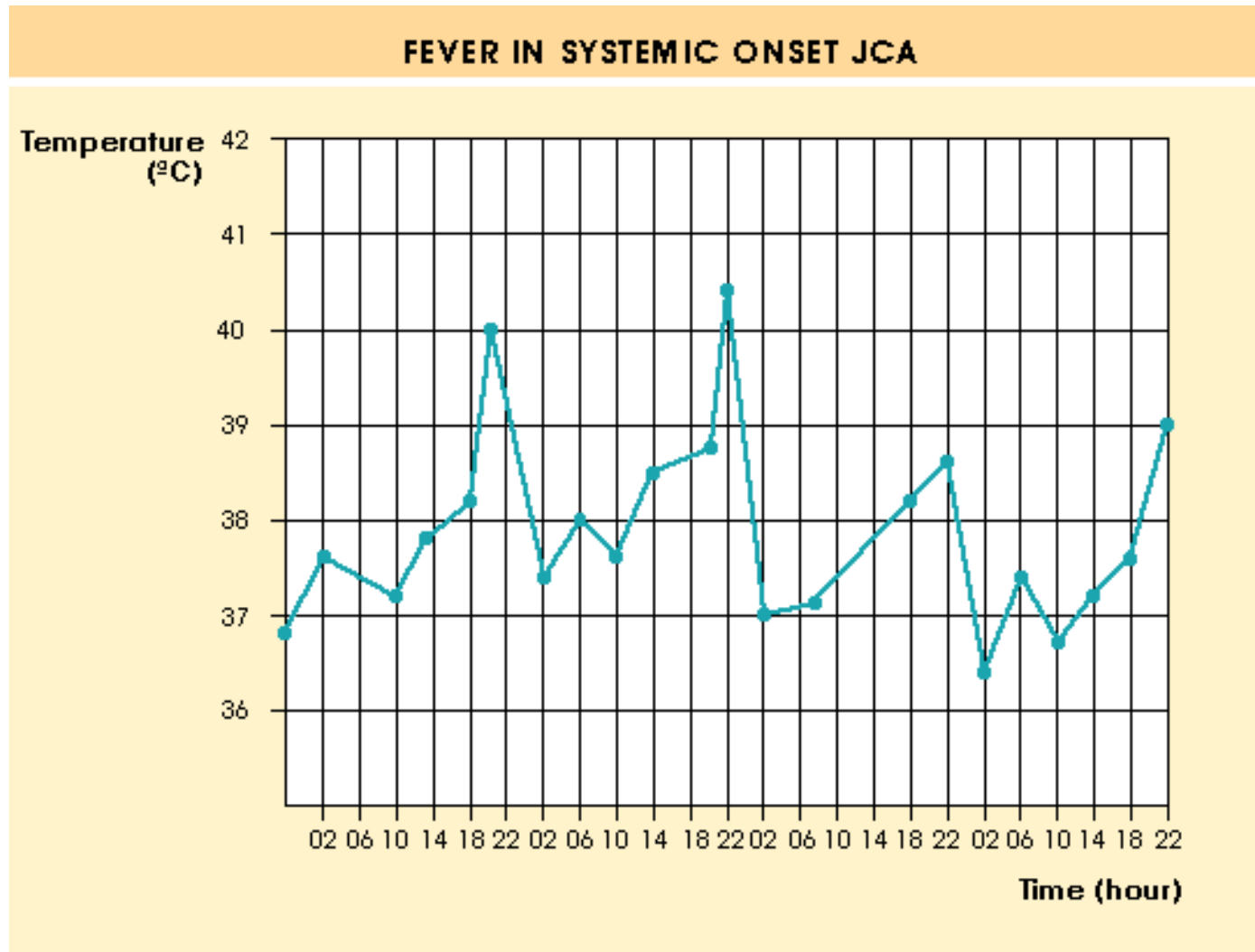
Extra-articular manifestations

- High spiking fevers
- Children are characteristically ill appearing when febrile, but seems to improve dramatically when their temperature is normal
- Macular, salmon pink rash
 - Brought out by heat and prominent during fever
- Hepatomegaly, splenomegaly and lymphadenopathy

Extra-articular manifestations

- Small pericardial effusion
- Chronic headaches and other neurological symptoms
- Pulmonary manifestations include pleural effusions

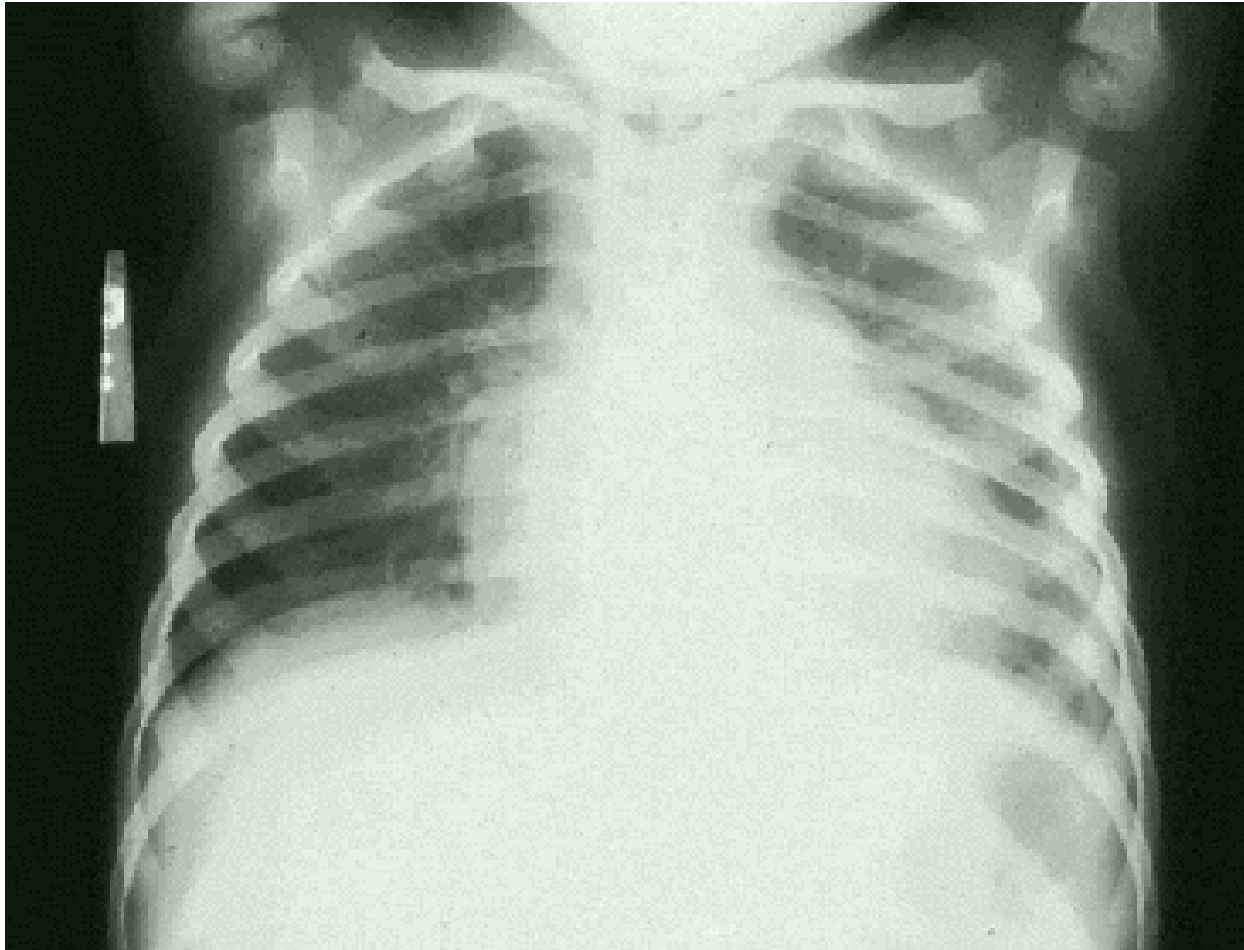
High swinging fever in systemic JIA





Rash of systemic onset JRA A salmon pink rash is characteristic of this JRA subtype. The rash is brought out by heat and often can be found in the axillae and around the waist, but may be present anywhere on the body. Courtesy of Tom Lehman, MD.

Pericardial effusion



Treatment of JIA

- NSAID
 - Naproxen, ibuprofen
- Steroids
 - Intra-articular injections are useful
 - Systemic steroids should be avoided
- Methotrexate
- Anti-TNF α , anti-IL6
- Autologous haematopoietic stem cell transplant

Physical and occupational therapy

- Maintenance of joint function
 - splints
 - ROM exercises (active/ passive)
 - posture/ prone-lying exercises
- Adequate rest/adequate exercise
 - balanced physical activity
 - selected peer sports (tricycle/ swimming)

Course and prognosis

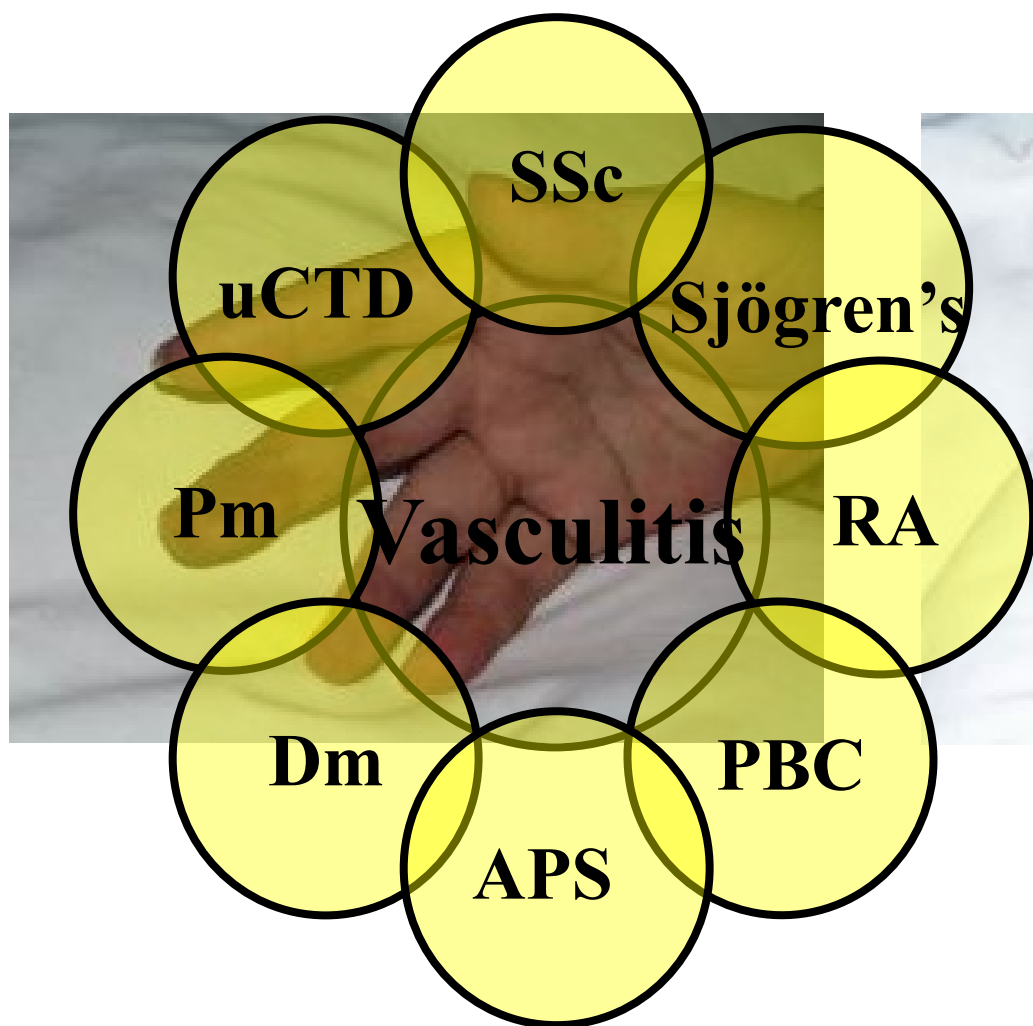
- Arthritis resolves in ~40-50% of children.
- If the child is well after 6 months, he/she is likely to remain well
- ~one-third of children with systemic JIA remain ill for a prolonged period of time – guarded prognosis

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- Sjögren's syndrome
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- **Primary vasculitides (optional)**

The following slides on vasculitis are not part of this lecture or other whole class sessions but students are welcome to refer to these materials during their revision

Vasculitis and connective tissue disorders



APS - Anti-phospholipid
antibody syndrome

Dm - Dermatomyositis

PBC - Primary biliary
cirrhosis

Pm - Polymyositis

RA - Rheumatoid arthritis

SSc - Systemic sclerosis

uCTD - Undifferentiated
connective tissue disorder

Vasculitic syndromes

- Primary vasculitis

- Giant cell arteritis
- Takayasu's arteritis
- Kawasaki disease
- Polyarteritis nodosa
- Eosinophilic granulomatosis with polyangiitis (Churg Strauss syndrome)
- Granulomatosis with polyangiitis (Wegener's granulomatosis)
- Microscopic angiitis
- IgA vasculitis (Henoch Schönlein purpura)
- Hypocomplementaemic urticarial vasculitis (anti-C1q vasculitis)

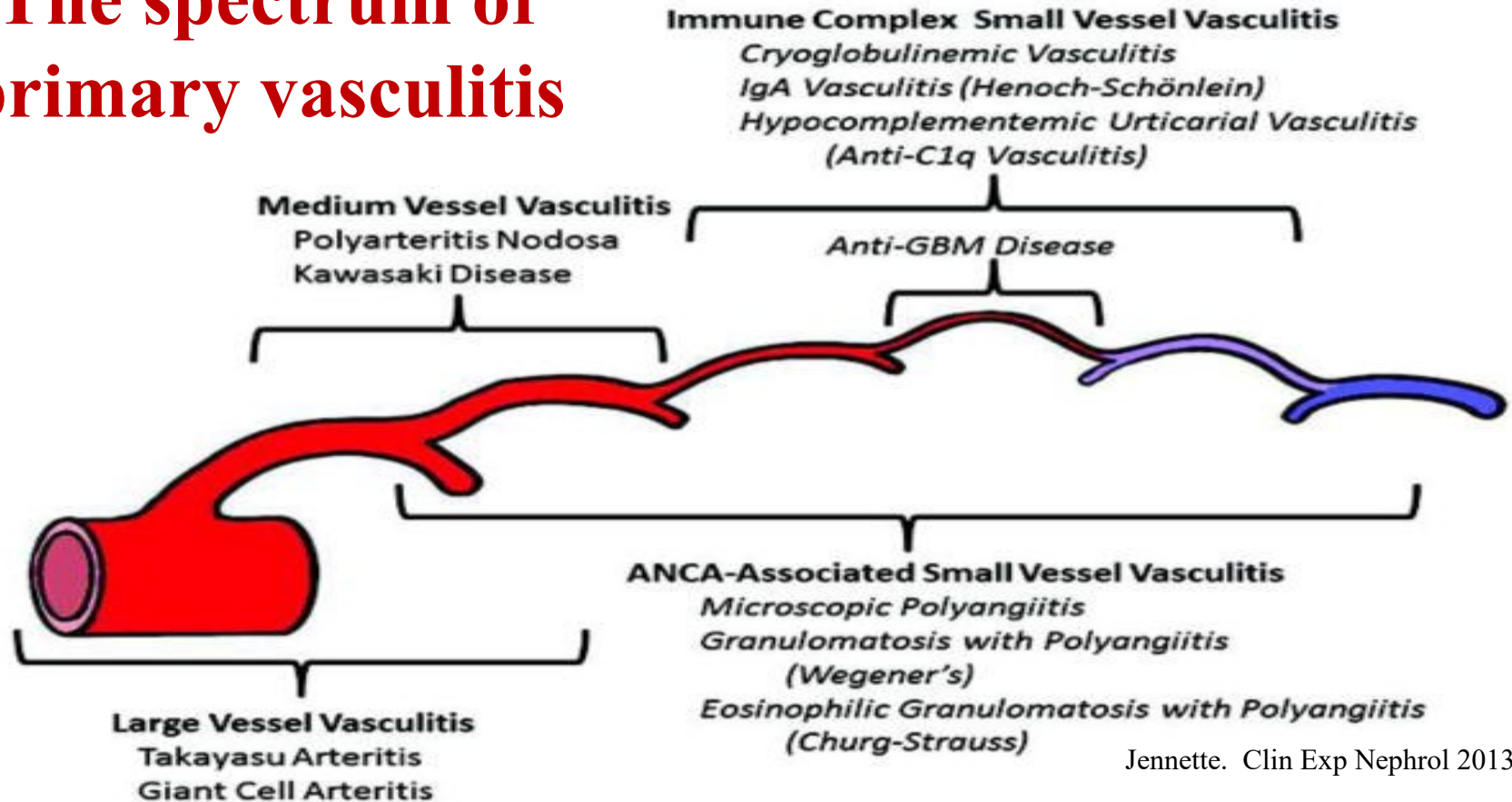
- Secondary vasculitis

- Systemic lupus erythematosus
- Rheumatoid arthritis
- Sjögren's syndrome
- Behcet's disease
- Essential mixed cryoglobulinaemia

Classification of primary vasculitis

- Large vessel disease
 - Giant cell arteritis
 - Takayasu's arteritis
- Medium / small vessel disease
 - Polyarteritis nodosa (PAN) (Viral hepatitis B related)
 - Kawasaki disease
 - Granulomatosis with polyangiitis (Wegener's granulomatosis) (ANCA related)
 - Eosinophilic granulomatosis with polyangiitis (Churg Strauss syndrome) (ANCA related)
 - Microscopic polyangiitis (ANCA related)
 - IgA vasculitis (Henoch Schönlein Purpura)

The spectrum of primary vasculitis



Jennette. Clin Exp Nephrol 2013; 603

Large vessel vasculitis

- Specific organ disease

Medium / small vessel vasculitis

- Multi-organ disease: Skin, renal, neurological and pulmonary
- Specific markers such as ANCA and eosinophilia

- Vessel wall destruction leads to perforation and haemorrhage into adjacent tissues
- Endothelial injury leads to thrombosis and ischaemia of dependent tissues
- Fever and other constitutional symptoms

Common clinical features of vasculitic syndromes



Livedo reticularis

Common clinical features of vasculitic syndromes



Nail fold erythema



Splinter haemorrhages



Palmar / finger pulp vasculitis

Common clinical features of vasculitic syndromes



Non-blanching palpable purpuric
lesions commonly found on
dependent areas / pressure sites

Common clinical features of vasculitic syndromes



Vascular insufficiency

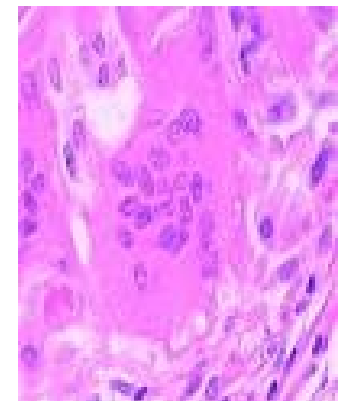
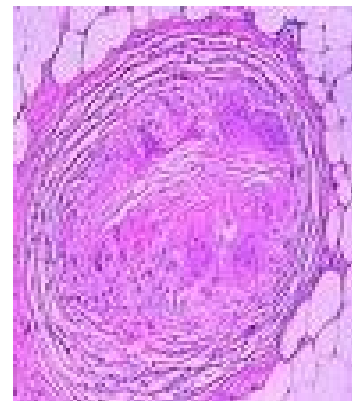
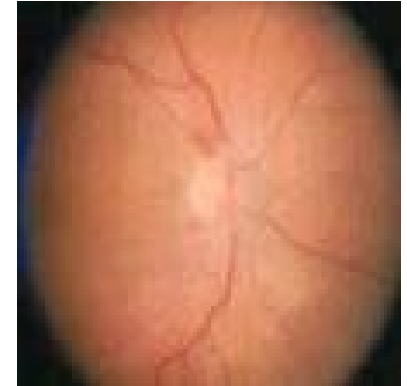
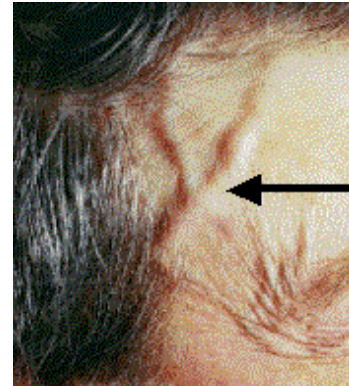
Common clinical features of vasculitic syndromes



Vasculitic ulcers commonly found in lower limbs
and pressure sites

Giant cell arteritis (Temporal arteritis)

- A granulomatous arteritis of the aorta and its major branches
- Commonest form of 1° vasculitis
 - ~20/100,000 person-years
 - Female : male = 2 : 1
 - Predominantly the elderly
- Clinical features
 - Fever and constitutional symptoms
 - Temporal headache with tender, non pulsatile temporal artery
 - Scalp tenderness
 - Jaw claudication
 - Loss of vision



Giant cell arteritis

- Investigations
 - ESR / PV
 - Temporal artery biopsy
 - Colour duplex US
 - Aortography if indicated - uncommon
- Treatment
 - High dose corticosteroids
 - Prednisolone 1-2 mg/kg/d
 - Methotrexate
 - Anti-IL6



GCA and PMR often co-exist and may represent clinical subsets of a single disease process. However, treatment of these conditions is not the same. PMR patients require lower doses of steroid

Takayasu's arteritis

- Females of child bearing age
- More common in Asians
- Symptoms occur due to stenosis of the aorta and its branches. Aortic arch and abdominal aorta most commonly affected
- Also known as
 - Pulseless disease
 - Aortic arch syndrome
 - Occlusive thrombo-aortopathy

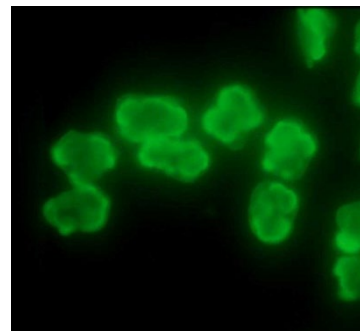


Takayasu's arteritis

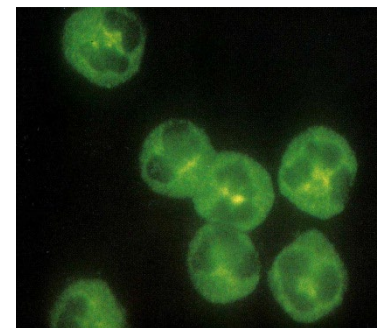
- Clinical features
 - Bruits (80%)
 - Claudication (70%)
 - Decreased pulses (60%)
 - Arthralgias (50%)
 - Asymmetric BP (50%)
 - Constitutional symptoms (40%)
 - Hypertension (30%)
- Investigations
 - Markers of inflammation
 - Aortography
 - MR angiography
- Treatment
 - High dose corticosteroids
 - Immunosuppressive drugs
 - Anti-IL6

ANCA related vasculitis

- Anti-neutrophil cytoplasmic antibodies
 - pANCA (p = perinuclear)
(Anti-myeloperoxidase [MPO] Ab)
 - cANCA (c = cytoplasmic)
(Anti-proteinase 3 [PR3] Ab)
 - Atypical ANCA (non-specific)



pANCA



cANCA

Disease entity	Anti-PR3	Anti-MPO
Polyarteritis nodosa	5%	15%
Granulomatosis with polyangiitis	85%	10%
Eosinophilic granulomatosis with polyangiitis	10%	66%
Microscopic polyangiitis	15-45%	45-80%

ANCA related vasculitis

- All are rare but potentially life-threatening systemic disorders
- Fever and constitutional complaints
- Multi-organ involvement of acute / sub-acute onset
 - Peripheral eosinophilia in some patients
 - Cutaneous vasculitis
 - Respiratory tract disease
 - Renal disease
 - Neurological disease
- Treatment involves high dose steroids and immunosuppressive drugs
- Rituximab (anti-CD20)

Polyarteritis and microscopic polyangiitis

Features	PAN	MPA
Renal involvement		
Vasculitis with infarcts / microaneurysms	Yes	No
Rapidly progressive glomerulonephritis	No	Yes
Lung involvement		
Pulmonary haemorrhage	No	Yes
Laboratory findings		
HBV-infection	Yes (50%)	No
pANCA	<20%	50-80%
Abnormal angiogram with microaneurysms	Yes	No
Relapses	Rare	Common

Eosinophilic granulomatosis polyangiitis and granulomatosis polyangiitis

Features	EGP	GP
ENT	Rhinitis, polyposis	Necrotising lesions
Allergy, bronchial asthma	Frequent	Similar to general population
Renal involvement	Uncommon	
Eosinophilia	>10% of peripheral white cells	Minimally elevated
ANCA	pANCA (anti-MPO) 66%	cANCA (anti-PR3) 85%
Histology	Eosinophilic necrotising granuloma	Necrotising epithelioid granuloma
Major cause of death	Cardiac	Pulmonary and renal

IgA vasculitis (Henoch Schönlein purpura)

- Hypersensitivity reaction to bacteria or viruses
- Occurs commonly in spring and after an URTI
- Children (2-10 yrs) are more commonly affected
- Adults may also be affected
- Classical triad
 - Acute onset / rapid resolution
 - Palpable purpura
 - Arthritis
 - Abdominal pain



TELL US WHAT
SC Whole Class Session
YOU THINK

THANK YOU!



**Look for today's teacher
and the others you may have missed.**

(HKU Portal login required)

Teacher: Prof CS Lau